ATRIA
INSTITUTE OF TECHNOLOGY
(AN AUTONOMOUS INSTITUTION)
BENGALURU

Atria's Societal Impactful STUDENT's Project Work Book

NAME OF SDG:03. GOOD HEALTH AND WELL BEING

TITTLE: WHEEZING PRE DETECTION AND ALERT SYSTEM

	TEAM LEAD's DETAILS BELOW:					
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Guide / Co-guide Name, Department		
Guide	Name:	Dept:
Co-Guide	Name:	Dept:

Course Outcomes (COs) for Semester 3:

At the end of the course, the student should be able to:

CO No	Course Outcome	Bloom's Level
CO1	Identify and analyze societal problems through stakeholder engagement and field-visits.	L4
CO2	Evaluate the feasibility and scope of identified problems to ensure alignment with Sustainable Development Goals (SDGs).	L5
CO3	Apply multidisciplinary knowledge to analyze problem context and feasibility.	L3
CO4	Define problem statements and objectives clearly with scope and constraints.	L3

CO5	Demonstrate effective teamwork, communication, and ethical understanding during the problem identification and definition process.	L3

Project Review Evaluation for Week

Evaluation of CO1: **Identify and analyze societal problems through stakeholder engagement and field-visits**

STUDENT # 1 NAME	USN	DEPT	SECTION
KISHOR S	1AT24UE033	ECM	A

Write your Problem Statement:

“Current medical devices for asthma and wheezing patients only detect breathing problems after the wheezing attack has already begun, which is often too late to prevent severe discomfort, panic, and emergency hospitalization. The critical gap is the lack of an affordable, user-friendly system that can continuously monitor breathing patterns and proactively predict an upcoming wheezing attack, thereby enabling timely intervention to save lives and reduce healthcare costs."

- **Step 1: Understanding the context - Empathize Stage**

1. **Who are the people or communities affected by this issue?**

The people affected by this issue are primarily individuals with asthma, COPD, and other respiratory problems who often experience undetected wheezing. Children, elderly patients, and those living in high-pollution areas are especially vulnerable. Caregivers, parents, and healthcare professionals also face challenges because they rely on late symptoms to respond. Communities in rural or low-resource areas are affected the most due to limited access to early-detection tools.

2. Where does this problem occur (geographic / institutional / social context)? This problem occurs in places with high air pollution such as urban areas, crowded neighborhoods, and poorly ventilated indoor spaces. It is also common in homes, schools, and hospitals where people with respiratory issues are frequently exposed to triggers.

3. When and how often does the problem happen?

This problem happens whenever individuals with asthma or respiratory issues are exposed to triggers like dust, pollution, cold air, or allergens. It occurs frequently during seasonal changes, pollution spikes, or physical activity when breathing becomes strained.

- **Step 2: Getting the stakeholder / end-user perspective**

4. What difficulties or pain points do people (end users) experience? People experience difficulty in detecting early wheezing, which leads to delayed treatment and sudden breathing discomfort. They also face anxiety, frequent hospital

visits, and challenges in managing symptoms without proper monitoring tools.

5. **What do users currently do to solve or manage this issue?**
   Users currently rely on inhalers, medications, and manual observation to manage wheezing symptoms. Many also visit hospitals or use basic breathing exercises, but these methods often fail to detect early warning signs.

- ## Step 3: Analyzing the root-cause of the problem

6. **What factors contribute to the issue (technical, human, environmental, systemic)?**

The issue is caused by environmental triggers like pollution, dust, and allergens, along with human factors such as poor awareness and delayed response to symptoms. Technical and systemic factors like lack of early-detection devices and limited access to healthcare also contribute to the problem.

7. **What would happen if we didn't solve it?**
   If we don't solve this issue, individuals may experience worsening wheezing that can escalate into severe asthma attacks or respiratory emergencies. This can lead to frequent hospitalizations, higher health risks, and a decline in overall quality of life.

8. **Why does this problem exist?**

This problem exists because most people lack proper tools or knowledge to detect early wheezing before it becomes severe. It is also made worse by environmental pollution and limited access to affordable respiratory monitoring systems.

- **Step 4: Clear Definition of the problem**

9. How will solving this problem create impact (social, environmental, economic)?

Solving this problem enables early prediction of wheezing attacks, reducing emergency cases and improving the quality of life for patients, especially those in vulnerable communities. It also lowers healthcare costs, minimizes work loss, and supports environmental awareness by linking respiratory issues with pollution exposure.

10. Is the problem statement focused on the user's need (not just a technical issue)?

Yes, the problem statement is focused on the user's need because it addresses the real difficulty patients face—unexpected breathing problems with no early warning. It prioritizes improving their safety, comfort, and quality of life rather than just describing a technical detection challenge.

11. Is the scope manageable with the available time and resources?

es, the project scope for a "Smart Wheezing Pre Detection and Alert System" appears manageable for the time and resources typical of an academic project. The focus is specific: continuously monitoring breathing patterns to predict attacks before they occur, which is a contained goal achievable with standard sensors and processing. Furthermore, the system is designed to be affordable and user-friendly, suggesting the use of practical, readily available components which aids in project execution within constraints.

- **Step 5: Validation of the problem with the end user**

12. Have you verified the problem with real users or data? Describe the interaction summary.

Yes, the problem was verified through real user interactions during field visits with individuals such as Mrs. Nandini (a homemaker), Mrs. Kumari (a classical dancer), and Dr. Sheela, a medical professional. Both patients reported experiencing sudden breathing difficulties and struggling with affordability of medicines, while the doctor confirmed the medical relevance and advised consulting pulmonologists for device validation. These interactions helped confirm that unexpected wheezing attacks, delayed detection, and cost barriers are genuine user needs that require a practical, affordable early-warning solution.

13. Do stakeholders agree this is a priority issue? Justify.

Yes, stakeholders likely agree this is a priority issue because the current medical system fails to prevent attacks, only detecting wheezing after it starts, which leads to severe discomfort and emergency hospitalizations. The project addresses this critical failure by aiming to predict attacks before they happen, which is viewed as an "urgent need". The solution is prioritized because it has a strong social impact, potentially saving lives and reducing healthcare costs by preventing emergencies.

14. Has this problem already been solved elsewhere — and can you learn from it?

The general problem of wheezing detection is partially solved by commercial and academic

devices that use sensors and Machine Learning (ML) to accurately detect wheezing in real-time. However, the project's critical focus is pre-detection—predicting an attack before it starts—which is a major unsolved challenge. Learning from existing systems involves studying how they analyze physiological and environmental data to identify the subtle, pre-symptomatic changes needed to build a successful predictive model.

Evaluation of CO1: Identify and analyze societal problems through stakeholder engagement and field-visits

STUDENT # 2 NAME	USN	DEPT	SECTION
SURAJ. T	1AT24CS225	CSE	D

Write your Problem Statement:

Current medical devices for asthma and wheezing patients only detect breathing problems after the wheezing attack has already begun, which is often too late to prevent severe discomfort, panic, and emergency hospitalization. The critical gap is the lack of an affordable, user-friendly system that can continuously monitor breathing patterns and proactively predict an upcoming wheezing attack, thereby enabling timely intervention to save lives and reduce healthcare costs."

- **Step 1: Understanding the context - Empathize Stage**

15. **Who are the people or communities affected by this issue? The people affected by this issue are primarily individuals with asthma, COPD, and other respiratory problems who often experience undetected wheezing. Children, elderly patients, and those living in high-pollution areas are especially vulnerable. Caregivers, parents, and healthcare professionals also face challenges because they rely on late symptoms to respond. Communities in rural or low-resource areas are affected the most due to limited access to early-detection tools.**

16. **Where does this problem occur (geographic / institutional / social context)? This problem occurs in places with high air pollution such as urban areas, crowded neighborhoods, and poorly ventilated indoor spaces. It is also common in homes, schools, and hospitals where people with respiratory issues are frequently exposed to triggers.**

17. **When and how often does the problem happen?**

This problem happens whenever individuals with asthma or respiratory issues are exposed to triggers like dust, pollution, cold air, or allergens. It occurs frequently during seasonal changes, pollution spikes, or physical activity when breathing becomes strained.

- **Step 2: Getting the stakeholder / end-user perspective**

18. What difficulties or pain points do people (end users) experience?

People experience difficulty in detecting early wheezing, which leads to delayed treatment and sudden breathing discomfort. They also face anxiety, frequent hospital visits, and challenges in managing symptoms without proper monitoring tools.

19. What do users currently do to solve or manage this issue?

Users currently rely on inhalers, medications, and manual observation to manage wheezing symptoms. Many also visit hospitals or use basic breathing exercises, but these methods often fail to detect early warning signs.

- **Step 3: Analyzing the root-cause of the problem**

20. What factors contribute to the issue (technical, human, environmental, systemic)?

The issue is caused by environmental triggers like pollution, dust, and allergens, along with human factors such as poor awareness and delayed response to symptoms. Technical and systemic factors like lack of early-detection devices and limited access to healthcare also contribute to the problem.

21. What would happen if we didn’t solve it?

If we don’t solve this issue, individuals may experience worsening wheezing that can escalate into severe asthma attacks or respiratory emergencies. This can lead to frequent hospitalizations, higher health risks, and a decline in overall quality of life.

22. Why does this problem exist? **This problem exists because most people lack proper tools or knowledge to detect early wheezing before it becomes severe. It is also made worse by environmental pollution and limited access to affordable respiratory monitoring systems**
• **Step 4: Clear Definition of the problem**
23. How will solving this problem create impact (social, environmental, economic)? **Solving this problem enables early prediction of wheezing attacks, reducing emergency cases and improving the quality of life for patients, especially those in vulnerable communities. It also lowers healthcare costs, minimizes work loss, and supports environmental awareness by linking respiratory issues with pollution exposure.**
24. Is the problem statement focused on the user's need (not just a technical issue)? **Yes, the problem statement is focused on the user's need because it addresses the real difficulty patients face—unexpected breathing problems with no early warning. It prioritizes improving their safety, comfort, and quality of life rather than just describing a technical detection challenge.**
25. Is the scope manageable with the available time and resources? **the project scope for a "Smart Wheezing Pre Detection and Alert System" appears manageable for the time and resources typical of an academic project. The focus is specific: continuously monitoring breathing patterns to predict attacks before they occur, which is a contained goal achievable with standard sensors and processing. Furthermore, the system**

is designed to be affordable and user-friendly, suggesting the use of practical, readily available components which aids in project execution within constraints

- **Step 5: Validation of the problem with the end user**

26. Have you verified the problem with real users or data? Describe the interaction summary.

Yes, the problem was verified through real user interactions during field visits with individuals such as Mrs. Nandini (a homemaker), Mrs. Kumari (a classical dancer), and Dr. Sheela, a medical professional. Both patients reported experiencing sudden breathing difficulties and struggling with affordability of medicines, while the doctor confirmed the medical relevance and advised consulting pulmonologists for device validation. These interactions helped confirm that unexpected wheezing attacks, delayed detection, and cost barriers are genuine user needs that require a practical, affordable early-warning solution.

27. Do stakeholders agree this is a priority issue? Justify.

Yes, stakeholders likely agree this is a priority issue because the current medical system fails to prevent attacks, only detecting wheezing after it starts, which leads to severe discomfort and emergency hospitalizations. The project addresses this critical failure by aiming to predict attacks before they happen, which is viewed as an "urgent need". The solution is prioritized because it has a strong social impact, potentially saving lives

and reducing healthcare costs by preventing emergencies.

28. Has this problem already been solved elsewhere — and can you learn from it?

The general problem of wheezing detection is partially solved by commercial and academic devices that use sensors and Machine Learning (ML) to accurately detect wheezing in real-time. However, the project's critical focus is pre-detection—predicting an attack before it starts—which is a major unsolved challenge. Learning from existing systems involves studying how they analyze physiological and environmental data to identify

Evaluation of CO1: Identify and analyze societal problems through stakeholder engagement and field-visits

STUDENT # 3 NAME	USN	DEPT	SECTION
RAMYA CHARITHA K A	IAT24AI082	AIML	B

Write your Problem Statement:

Current medical devices for asthma and wheezing patients only detect breathing problems after the wheezing attack has already begun, which is often too late to prevent severe discomfort, panic, and emergency hospitalization. The critical gap is the lack of an affordable, user-friendly system that can continuously monitor breathing patterns and

proactively predict an upcoming wheezing attack, thereby enabling timely intervention to save lives and reduce healthcare costs."

- **Step 1: Understanding the context - Empathize Stage**

29. **Who are the people or communities affected by this issue? The people affected by this issue are primarily individuals with asthma, COPD, and other respiratory problems who often experience undetected wheezing. Children, elderly patients, and those living in high-pollution areas are especially vulnerable. Caregivers, parents, and healthcare professionals also face challenges because they rely on late symptoms to respond. Communities in rural or low-resource areas are affected the most due to limited access to early-detection tools.**

30. **Where does this problem occur (geographic / institutional / social context)? This problem occurs in places with high air pollution such as urban areas, crowded neighborhoods, and poorly ventilated indoor spaces. It is also common in homes, schools, and hospitals where people with respiratory issues are frequently exposed to triggers.**

31. When and how often does the problem happen?

This problem happens whenever individuals with asthma or respiratory issues are exposed to triggers like dust, pollution, cold air, or allergens. It occurs frequently during seasonal changes, pollution spikes, or physical activity when breathing becomes strained.

- **Step 2: Getting the stakeholder / end-user perspective**

32. What difficulties or pain points do people (end users) experience? People experience difficulty in detecting early wheezing, which leads to delayed treatment and sudden breathing discomfort. They also face anxiety, frequent hospital visits, and challenges in managing symptoms without proper monitoring tools.

33. What do users currently do to solve or manage this issue?

Users currently rely on inhalers, medications, and manual observation to manage wheezing symptoms. Many also visit hospitals or use basic breathing exercises, but these methods often fail to detect early warning signs.

- **Step 3: Analyzing the root-cause of the problem**

34. What factors contribute to the issue (technical, human, environmental, systemic)?

The issue is caused by environmental triggers like pollution, dust, and allergens, along with human factors such as poor awareness and delayed response to symptoms. Technical and

systemic factors like lack of early-detection devices and limited access to healthcare also contribute to the problem

35. What would happen if we didn't solve it?

If we don't solve this issue, individuals may experience worsening wheezing that can escalate into severe asthma attacks or respiratory emergencies. This can lead to frequent hospitalizations, higher health risks, and a decline in overall quality of life.

36. Why does this problem exist?

This problem exists because most people lack proper tools or knowledge to detect early wheezing before it becomes severe. It is also made worse by environmental pollution and limited access to affordable respiratory monitoring systems.

- **Step 4: Clear Definition of the problem**

37. How will solving this problem create impact (social, environmental, economic)?

Solving this problem enables early prediction of wheezing attacks, reducing emergency cases and improving the quality of life for patients, especially those in vulnerable communities. It also lowers healthcare costs, minimizes work loss, and supports environmental awareness by linking respiratory issues with pollution exposure.

38. Is the problem statement focused on the user's need (not just a technical issue)?

Yes, the problem statement is focused on the user's need because it addresses the real difficulty patients face—unexpected breathing problems with no early warning. It prioritizes improving their safety, comfort, and quality of life rather than just describing a technical detection challenge.

39. Is the scope manageable with the available time and resources?

the project scope for a "Smart Wheezing Pre Detection and Alert System" appears manageable for the time and resources typical of an academic project. The focus is specific: continuously monitoring breathing patterns to predict attacks before they occur, which is a contained goal achievable with standard sensors and processing. Furthermore, the system is designed to be affordable and user-friendly, suggesting the use of practical, readily available components which aids in project execution within constraints.

- **Step 5: Validation of the problem with the end user**

40. Have you verified the problem with real users or data? Describe the interaction summary.

Yes, the problem was verified through real user interactions during field visits with individuals such as Mrs. Nandini (a homemaker), Mrs. Kumari (a classical dancer), and Dr. Sheela, a medical professional. Both patients reported experiencing sudden breathing difficulties and struggling with affordability of medicines, while the doctor confirmed the medical relevance and advised consulting pulmonologists for device

validation. These interactions helped confirm that unexpected wheezing attacks, delayed detection, and cost barriers are genuine user needs that require a practical, affordable early-warning solution.

41. Do stakeholders agree this is a priority issue? Justify.

Yes, stakeholders likely agree this is a priority issue because the current medical system fails to prevent attacks, only detecting wheezing after it starts, which leads to severe discomfort and emergency hospitalizations. The project addresses this critical failure by aiming to predict attacks before they happen, which is viewed as an "urgent need". The solution is prioritized because it has a strong social impact, potentially saving lives and reducing healthcare costs by preventing emergencies.

42. Has this problem already been solved elsewhere — and can you learn from it?

Evaluation of CO1: Identify and analyze societal problems through stakeholder engagement and field-visits

STUDENT # 4 NAME	USN	DEPT	SECTION

TANUSHRI M P	**1AT24CS236**	**CSE**	**D**

Write your Problem Statement:

Current medical devices for asthma and wheezing patients only detect breathing problems after the wheezing attack has already begun, which is often too late to prevent severe discomfort, panic, and emergency hospitalization. The critical gap is the lack of an affordable, user-friendly system that can continuously monitor breathing patterns and proactively predict an upcoming wheezing attack, thereby enabling timely intervention to save lives and reduce healthcare costs."

- **Step 1: Understanding the context - Empathize Stage**

43. Who are the people or communities affected by this issue?
The people affected by this issue are primarily individuals with asthma, COPD, and other respiratory problems who often experience undetected wheezing. Children, elderly patients, and those living in high-pollution areas are especially vulnerable. Caregivers, parents, and healthcare professionals also face challenges because they rely on late symptoms to respond. Communities in rural or low-resource areas are affected the most due to limited access to early-detection tools.

44. Where does this problem occur (geographic / institutional / social context)? This problem occurs in places with high air pollution such as urban areas, crowded neighborhoods, and poorly ventilated indoor spaces. It is also common in homes, schools, and hospitals where people with respiratory issues are frequently exposed to triggers.

45. When and how often does the problem happen?

This problem happens whenever individuals with asthma or respiratory issues are exposed to triggers like dust, pollution, cold air, or allergens. It occurs frequently during seasonal changes, pollution spikes, or physical activity when breathing becomes strained

- **Step 2: Getting the stakeholder / end-user perspective**

46. What difficulties or pain points do people (end users) experience? People experience difficulty in detecting early wheezing, which leads to delayed treatment and sudden breathing discomfort. They also face anxiety, frequent hospital visits, and challenges in managing symptoms without proper monitoring tools.

47. What do users currently do to solve or manage this issue?

Users currently rely on inhalers, medications, and manual observation to manage wheezing symptoms. Many also visit hospitals or use basic breathing exercises, but these methods often fail to detect early warning signs.

- ## Step 3: Analyzing the root-cause of the problem

48. **What factors contribute to the issue (technical, human, environmental, systemic)?**

The issue is caused by environmental triggers like pollution, dust, and allergens, along with human factors such as poor awareness and delayed response to symptoms. Technical and systemic factors like lack of early-detection devices and limited access to healthcare also contribute to the problem

49. **What would happen if we didn't solve it?**

    If we don't solve this issue, individuals may experience worsening wheezing that can escalate into severe asthma attacks or respiratory emergencies. This can lead to frequent hospitalizations, higher health risks, and a decline in overall quality of life.

50. **Why does this problem exist?**

This problem exists because most people lack proper tools or knowledge to detect early wheezing before it becomes severe. It is also made worse by environmental pollution and limited access to affordable respiratory monitoring systems.

- **Step 4: Clear Definition of the problem**

51.How will solving this problem create impact (social, environmental, economic)?

Solving this problem enables early prediction of wheezing attacks, reducing emergency cases and improving the quality of life for patients, especially those in vulnerable communities. It also lowers healthcare costs, minimizes work loss, and supports environmental awareness by linking respiratory issues with pollution exposure.

52. Is the problem statement focused on the user's need (not just a technical issue)?

Yes, the problem statement is focused on the user's need because it addresses the real difficulty patients face—unexpected breathing problems with no early warning. It prioritizes improving their safety, comfort, and quality of life rather than just describing a technical detection challenge.

53. Is the scope manageable with the available time and resources?

the project scope for a "Smart Wheezing Pre Detection and Alert System" appears manageable for the time and resources typical of an academic project. The focus is specific: continuously monitoring breathing patterns to predict attacks before they occur, which is a contained goal achievable with standard sensors and processing. Furthermore, the system is designed to be affordable and user-friendly, suggesting the use of practical, readily available components which aids in project execution within constraints.

- **Step 5: Validation of the problem with the end user**

54. Have you verified the problem with real users or data? Describe the interaction summary.

Yes, the problem was verified through real user interactions during field visits with individuals such as Mrs. Nandini (a homemaker), Mrs. Kumari (a classical dancer), and Dr. Sheela, a medical professional. Both patients reported experiencing sudden breathing difficulties and struggling with affordability of medicines, while the doctor confirmed the medical relevance and advised consulting pulmonologists for device validation. These interactions helped confirm that unexpected wheezing attacks, delayed detection, and cost barriers are genuine user needs that require a practical, affordable early-warning solution.

55. Do stakeholders agree this is a priority issue? Justify.

Yes, stakeholders likely agree this is a priority issue because the current medical system fails to prevent attacks, only detecting wheezing after it starts, which leads to severe discomfort and emergency hospitalizations. The project addresses this critical failure by aiming to predict attacks before they happen, which is viewed as an "urgent need". The solution is prioritized because it has a strong social impact, potentially saving lives and reducing healthcare costs by preventing emergencies.

56. Has this problem already been solved elsewhere — and can you learn from it?

The general problem of wheezing detection is partially solved by commercial and academic devices that use sensors and Machine Learning (ML) to accurately detect wheezing in real-

time. However, the project's critical focus is pre-detection—predicting an attack before it starts—which is a major unsolved challenge. Learning from existing systems involves studying how they analyze physiological and environmental data to identify

Evaluation of CO1: **Identify and analyze societal problems through stakeholder engagement and field-visits**

STUDENT # 5 NAME	USN	DEPT	SECTION
MANOHAR.N P	IAT24IS110	ISE	D

Write your Problem Statement:

Current medical devices for asthma and wheezing patients only detect breathing problems after the wheezing attack has already begun, which is often too late to prevent severe discomfort, panic, and emergency hospitalization. The critical gap is the lack of an affordable, user-friendly system that can continuously monitor breathing patterns and proactively predict an upcoming wheezing attack, thereby enabling timely intervention to save lives and reduce healthcare costs."

- **Step 1: Understanding the context - Empathize Stage**

57. Who are the people or communities affected by this issue?
The people affected by this issue are primarily individuals with asthma, COPD, and other respiratory problems who often experience undetected wheezing. Children, elderly patients, and those living in high-pollution areas are especially vulnerable. Caregivers, parents, and healthcare professionals also face challenges because they rely on late symptoms to respond. Communities in rural or low-resource areas are affected the most due to limited access to early-detection tools.

58. Where does this problem occur (geographic / institutional / social context)?
This problem occurs in places with high air pollution such as urban areas, crowded neighborhoods, and poorly ventilated indoor spaces. It is also common in homes, schools, and hospitals where people with respiratory issues are frequently exposed to triggers.

59. When and how often does the problem happen?
This problem happens whenever individuals with asthma or respiratory issues are exposed to triggers like dust, pollution, cold air, or allergens. It occurs frequently during seasonal changes, pollution spikes, or physical activity when breathing becomes strained

- **Step 2: Getting the stakeholder / end-user perspective**

60. What difficulties or pain points do people (end users) experience?
People experience difficulty in detecting early wheezing, which leads to delayed

treatment and sudden breathing discomfort. They also face anxiety, frequent hospital visits, and challenges in managing symptoms without proper monitoring tools.
61. What do users currently do to solve or manage this issue? Users currently rely on inhalers, medications, and manual observation to manage wheezing symptoms. Many also visit hospitals or use basic breathing exercises, but these methods often fail to detect early warning signs.
• **Step 3: Analyzing the root-cause of the problem**
62. What factors contribute to the issue (technical, human, environmental, systemic)? The issue is caused by environmental triggers like pollution, dust, and allergens, along with human factors such as poor awareness and delayed response to symptoms. Technical and systemic factors like lack of early-detection devices and limited access to healthcare also contribute to the problem Use of unreliable medicines improper waste disposal.
63. What would happen if we didn't solve it? If we don't solve this issue, individuals may experience worsening wheezing that can escalate into severe asthma attacks or respiratory emergencies. This can lead to frequent hospitalizations, higher health risks, and a decline in overall quality of life.
64. Why does this problem exist? This problem exists because most people lack proper tools or knowledge to detect early wheezing before it becomes severe. It is also made worse by environmental pollution and

limited access to affordable respiratory monitoring systems.

- ## Step 4: Clear Definition of the problem

65. How will solving this problem create impact (social, environmental, economic)?

Solving this problem enables early prediction of wheezing attacks, reducing emergency cases and improving the quality of life for patients, especially those in vulnerable communities. It also lowers healthcare costs, minimizes work loss, and supports environmental awareness by linking respiratory issues with pollution exposure.

66. Is the problem statement focused on the user’s need (not just a technical issue)?

Yes, the problem statement is focused on the user’s need because it addresses the real difficulty patients face—unexpected breathing problems with no early warning. It prioritizes improving their safety, comfort, and quality of life rather than just describing a technical detection challenge.

67. Is the scope manageable with the available time and resources?

the project scope for a "Smart Wheezing Pre Detection and Alert System" appears manageable for the time and resources typical of an academic project. The focus is specific: continuously monitoring breathing patterns to predict attacks before they occur, which is a contained goal achievable with standard sensors and processing. Furthermore, the system is designed to be affordable and user-friendly, suggesting the use of practical, readily available components which aids in project execution within constraints.

- **Step 5: Validation of the problem with the end user**

68. Have you verified the problem with real users or data? Describe the interaction summary.

Yes, the problem was verified through real user interactions during field visits with individuals such as Mrs. Nandini (a homemaker), Mrs. Kumari (a classical dancer), and Dr. Sheela, a medical professional. Both patients reported experiencing sudden breathing difficulties and struggling with affordability of medicines, while the doctor confirmed the medical relevance and advised consulting pulmonologists for device validation. These interactions helped confirm that unexpected wheezing attacks, delayed detection, and cost barriers are genuine user needs that require a practical, affordable early-warning solution.

69. Do stakeholders agree this is a priority issue? Justify.

Yes, stakeholders likely agree this is a priority issue because the current medical system fails to prevent attacks, only detecting wheezing after it starts, which leads to severe discomfort and emergency hospitalizations. The project addresses this critical failure by aiming to predict attacks before they happen, which is viewed as an "urgent need". The solution is prioritized because it has a strong social impact, potentially saving lives and reducing healthcare costs by preventing emergencies.

70. Has this problem already been solved elsewhere — and can you learn from it?

The general problem of wheezing detection is partially solved by commercial and academic devices that use sensors and Machine Learning (ML) to accurately detect wheezing in real-

time. However, the project's critical focus is pre-detection—predicting an attack before it starts—which is a major unsolved challenge. Learning from existing systems involves studying how they analyze physiological and environmental data to identify

Evaluation of CO1: Identify and analyze societal problems through stakeholder engagement and field-visits

STUDENT # 6 NAME	USN	DEPT	SECTION
MAHALAKSHMI. K	1AT24UE036	ECM	A

Write your Problem Statement:

Current medical devices for asthma and wheezing patients only detect breathing problems after the wheezing attack has already begun, which is often too late to prevent severe discomfort, panic, and emergency hospitalization. The critical gap is the lack of an affordable, user-friendly system that can continuously monitor breathing patterns and proactively predict an upcoming wheezing attack, thereby enabling timely intervention to save lives and reduce healthcare costs."

- **Step 1: Understanding the context - Empathize Stage**

71. Who are the people or communities affected by this issue?
The people affected by this issue are primarily individuals with asthma, COPD, and other respiratory problems who often experience undetected wheezing. Children, elderly patients, and those living in high-pollution areas are especially vulnerable. Caregivers, parents, and healthcare professionals also face challenges because they rely on late symptoms to respond. Communities in rural or low-resource areas are affected the most due to limited access to early-detection tools.

72. Where does this problem occur (geographic / institutional / social context)?
This problem occurs in places with high air pollution such as urban areas, crowded neighborhoods, and poorly ventilated indoor spaces. It is also common in homes, schools, and hospitals where people with respiratory issues are frequently exposed to triggers.

73. When and how often does the problem happen?
This problem happens whenever individuals with asthma or respiratory issues are exposed to triggers like dust, pollution, cold air, or allergens. It occurs frequently during seasonal changes, pollution spikes, or physical activity when breathing becomes strained

- **Step 2: Getting the stakeholder / end-user perspective**

74. What difficulties or pain points do people (end users) experience?

People experience difficulty in detecting early wheezing, which leads to delayed treatment and sudden breathing discomfort. They also face anxiety, frequent hospital visits, and challenges in managing symptoms without proper monitoring tools.

75. What do users currently do to solve or manage this issue?

Users currently rely on inhalers, medications, and manual observation to manage wheezing symptoms. Many also visit hospitals or use basic breathing exercises, but these methods often fail to detect early warning signs.

They rely on local shop advice, YouTube tutorials, or random medicines, which often fail to solve the real cause.

- **Step 3: Analyzing the root-cause of the problem**

76. What factors contribute to the issue (technical, human, environmental, systemic)?

The issue is caused by environmental triggers like pollution, dust, and allergens, along with human factors such as poor awareness and delayed response to symptoms. Technical and systemic factors like lack of early-detection devices and limited access to healthcare also contribute to the problem

Poor water quality management, lack of nitrogen cycle knowledge, and unverified information sources are the main causes.

77. What would happen if we didn't solve it?

If we don't solve this issue, individuals may experience worsening wheezing that can escalate into severe asthma attacks or respiratory emergencies. This can lead to frequent hospitalizations, higher health risks, and a decline in overall quality of life.

78. Why does this problem exist?

This problem exists because most people lack proper tools or knowledge to detect early wheezing before it becomes severe. It is also made worse by environmental pollution and limited access to affordable respiratory monitoring systems.

- **Step 4: Clear Definition of the problem**

79. How will solving this problem create impact (social, environmental, economic)?

Solving this problem enables early prediction of wheezing attacks, reducing emergency cases and improving the quality of life for patients, especially those in vulnerable communities. It also lowers healthcare costs, minimizes work loss, and supports environmental awareness by linking respiratory issues with pollution exposure.

It promotes responsible fishkeeping, reduces water waste, and saves money for owners.

80. Is the problem statement focused on the user's need (not just a technical issue)?

Yes, the problem statement is focused on the user's need because it addresses the real difficulty patients face—unexpected breathing problems with no early warning. It prioritizes improving their safety, comfort, and quality of life rather than just describing a technical detection challenge.

81. Is the scope manageable with the available time and resources?

the project scope for a "Smart Wheezing Pre Detection and Alert System" appears manageable for the time and resources typical of an academic project. The focus is specific: continuously monitoring breathing patterns to predict attacks before they occur, which is a contained goal achievable with standard sensors and processing. Furthermore, the system is designed to be affordable and user-friendly, suggesting the use of practical, readily available components which aids in project execution within constraints.

- **Step 5: Validation of the problem with the end user**

82. Have you verified the problem with real users or data? Describe the interaction summary.

Yes, the problem was verified through real user interactions during field visits with individuals such as Mrs. Nandini (a homemaker), Mrs. Kumari (a classical dancer), and Dr. Sheela, a medical professional. Both patients reported experiencing sudden breathing difficulties and struggling with affordability of medicines, while the doctor confirmed the medical relevance and advised consulting pulmonologists for device validation. These interactions helped confirm that unexpected wheezing attacks, delayed detection, and cost barriers are genuine user needs that require a practical,

affordable early-warning solution.
83. Do stakeholders agree this is a priority issue? Justify. **Yes, stakeholders likely agree this is a priority issue because the current medical system fails to prevent attacks, only detecting wheezing after it starts, which leads to severe discomfort and emergency hospitalizations. The project addresses this critical failure by aiming to predict attacks before they happen, which is viewed as an "urgent need". The solution is prioritized because it has a strong social impact, potentially saving lives and reducing healthcare costs by preventing emergencies.**
84. Has this problem already been solved elsewhere — and can you learn from it? **The general problem of wheezing detection is partially solved by commercial and academic devices that use sensors and Machine Learning (ML) to accurately detect wheezing in real-time. However, the project's critical focus is pre-detection—predicting an attack before it starts—which is a major unsolved challenge. Learning from existing systems involves studying how they analyze physiological and environmental data to identify**